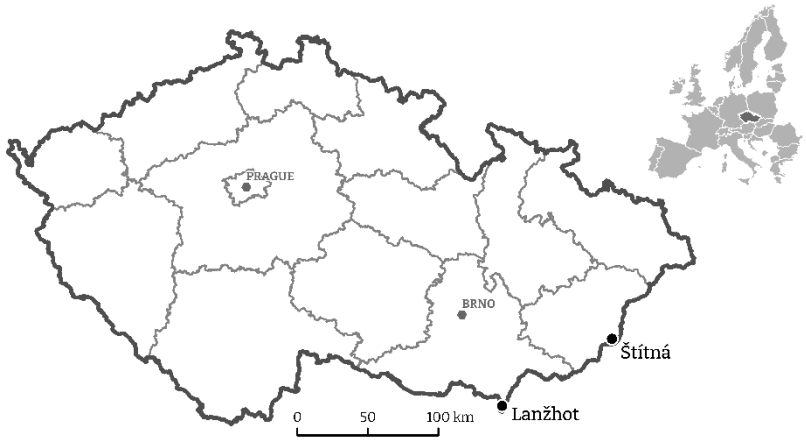
Plant functional traits and corresponding satellite multispectral data for two sites in the Czech Republic

This data set contains measurements of plant functional traits and corresponding satellite multispectral data for two sites in the Czech Republic – Lanžhot and Štítná throughout multiple campaigns. Plant functional traits were collected from representative trees selected at both study sites. At the Lanžhot site, a total of 16 trees of the following species were sampled: Quercus robur (4 trees), Quercus cerris L. (2 trees), Carpinus betulus (4 trees), Fraxinus angustifolia (4 trees), Tilia cordata (2 trees). At Štítná, a total of 10 Fagus sylvatica trees were sampled. Field campaigns were timed to cover different phenological stages of deciduous tree leaf development, from fresh leaf emergence, through full leaf development at the peak of the growing season, to autumn leaf senescence. In total, six field campaigns at Lanžhot during 2019 and 2020 and five field campaigns at Štítná during 2020 and 2021 were conducted.



Map of Czech Republic depicting the two sites, Lanžhot and Štítná.

The data set has following columns:

- **id**: unique identifier
- **tree_id**: unique identifier of the sampled tree
- **tree_species**: species of the sampled tree
- **date**: date of campaign (dd.mm.yyyy)
- **Car [µg cm^{-2}]:** carotenoid content
- **Car/Cab [-]:** carotenoid content/chlorophyll content
- **Cab [µg cm^{-2}]:** chlorophyll content
- **SLA [cm^2 g^{-1}]:** specific leaf area
- **Cw [%]:** water content
- **EWT [mg cm^{-2}]:** equivalent water thickness
- **S2_B02 – S2_B12:** Sentinel-2 reflectance
- **HLS_B02 – HLS_B07:** Harmonized Landsat Sentinel-2 reflectance